

Question

Fluorine gas reacts with hydrazoic acid to produce 3 products: a certain pseudohalogen-halogen interhalide, a gas, and a common salt. Which of the following statements is correct:

- A. The generated pseudohalogen-halogen interhalide is N_3F_3
- B. Gaseous product is ammonia.
- C. The stoichiometric ratio of fluorine gas to hydrazoic acid in this reaction may be 1:4
- D. The gaseous product is nitrogen trifluoride.
- E. For every mole of fluorine gas consumed in the reaction, 6 moles of gas may be produced.
- F. The anion in the salt produced by the reaction is N_3^-
- G. All other options are incorrect.

Answer

Correct Answer: **C**

Detailed Explanation

The only halogen in the question is fluorine, and the only pseudohalogen is azide, so the pseudohalogen-halogen interhalogen compound is fluorine azide.

CHECKPOINT

1 PTS

The pseudohalogen-halogen interhalogen compound is FN_3

Azide is oxidized to nitrogen gas by strongly oxidizing fluorine, so the gas is nitrogen gas.

CHECKPOINT

1 PTS

The gaseous product is N_2

The hydrogen element can only exist as ammonium ion, and the anion is the reduction product fluoride ion, so the salt is ammonium fluoride.

CHECKPOINT

1 PTS

The salt product is NH_4F

Therefore, options A, B, and D are all incorrect. Next, balance the equation. There are two reasonable equations:



CHECKPOINT

2 PTS



Fluorine azide is also a gas, so the stoichiometric ratio of fluorine gas to gas products is between 1:2 and 1:5.

CHECKPOINT

1 PTS

The stoichiometric ratio of fluorine gas to the produced gases is 1:2 to 1:5

E is incorrect.

The stoichiometric ratio of fluorine gas to hydrazoic acid is between 1:2 and 1:4.

CHECKPOINT

1 PTS

The stoichiometric ratio of fluorine gas to hydrazoic acid is 1:2 to 1:4

C is correct.